Role of species composition in malaria transmission by the *Anopheles funestus* group (Diptera: Culicidae) in Ghana

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ABSTRACT: Malaria remains a public health problem in Ghana, with *Anopheles gambiae* and *Anopheles funestus* as the predominant vectors. While much information exists on the species composition of *An. gambiae*, very little exists for *An. funestus*. This study was carried out to determine the species composition of *An. funestus* Giles populations from three ecological areas in Ghana and investigate their role in malaria transmission. Mosquitoes were collected using human landing and pyrethrum spray methods. A total of 10,254 *Anopheles* individuals were collected, out of which *An. funestus* constituted 53.6% (5,496). *An. funestus sensu stricto* (s.s.) and *Anopheles lessoni* were identified as the only members of the *An. funestus* group in all three ecological areas. All 62 sporozoite positive specimens that were identified as *An. funestus s.s.* were highly anthropophilic with a human blood index in the range of 80-96%, whereas more than 83% of the *An. leesoni* had fed on either bovine, goat, or sheep. Malaria transmission was higher in the Sahel savannah area than the rest of the ecological zones, with *An. funestus* s.s. being implicated as a vector of malaria in all ecological zones. *Anopheles leesoni* occurred in all the ecological areas but played no role in malaria transmission. The study established the importance of *An. funestus* s.s. in malaria transmission in Ghana. ***Journal of Vector Ecology* 38 (1): 105-110. 2013.**

Keyword Index: Mosquitoes, *Anopheles funestus*, ecology, malaria, transmission, Ghana.

INTRODUCTION

Anopheles funestus Giles is one of the two most important and widely distributed vectors of malaria in Africa (Gillies and de Meillon 1968). In some cases, its potential to transmit *Plasmodium falciparum* far outstrips that of *Anopheles gambiae* (Fontenille et al. 1997). It is also responsible for malaria epidemics in some areas (Fontenille et al. 1997). The species belongs to a group of nine morphologically similar but genetically different species (Gillies and Coetzee 1987). Recent findings have discovered '*An. rivulorum*' species as a potential tenth species in the *An. funestus* group (Cohuet et al. 2003). *An. funestus* sensu stricto (s.s) play a major role in malaria transmission in many areas (Gillies and Coetzee 1987, Fontenille et al. 1997, Costantini et al. 1999, Cohuet et al. 2004) and to a lesser extent *An. rivulorum* Leeson Evans (Gillies and Coetzee 1987, Wilkes et al. 1996, Kawada et al. 2012) The other members of the group are mainly zoophilic (Gillies and Coetzee 1987). *An. funestus* group may be genetically differentiated into locally adapted populations throughout its range in tropical Africa with different vectorial and behavioral capacities (Boccolini et al. 1994, Lochoarn et al. 1998, Costantini et al. 1999). It is therefore important that members of the group are accurately identified so as to be able to understand their role in malaria disease transmission patterns and hence formulate evidence-based control strategies to reduce malaria transmission. *An. gambiae* Giles and *An. funestus* Giles are widely distributed in Ghana and have been implicated as the major vectors of malaria in many areas of the country (Appawu 2001a, 2004). Much information is available on species composition and the vectorial role of the various species of *An. gambiae* complex (Appawu et al. 2001a, 2004, Yawson et al. 2004) in the country. *Anopheles funestus* has also been implicated as a vector of lymphatic filariasis in the coastal (Dunyo 1996) irrigated community (Dzodzomenyo et al. 1999) and in northern parts of Ghana (Appawu et al. 2001b). However, there is very little information on the species composition of *An. funestus* group across the different ecological areas in Ghana. Malaria vector control in the country will require more detailed knowledge of the genetic heterogeneities within populations of *An. funestus* and their effects on malaria transmission. The study determined the *An. funestus* group species composition from three ecological areas in Ghana and their biting and feeding behavior as well as the infection rates and entomological inoculation rates in order to better understand their epidemiologic role in malaria transmission.

MATERIALS AND METHODS

Mosquito collection sites and sampling methods

Mosquitoes were collected in three ecological areas: Navrongo, in the Sahel savannah belt (10º 30' N, 1º 30' W), Ejura, in a forest (06º 40' N, 09º 8.72' W), and Dodowa, in the coastal forest belt (05º 52.673' N, 000º 06.365' W). The average annual rainfall in the Sahel savannah area is 850 mm, most of which occurs in the wet months of May-September with the rest of the year being relatively dry. A large reservoir in the middle of the district provides water for irrigation. The vegetation of the forest site is mainly deciduous and the inhabitants are mostly crop farmers and small scale traders. The area is characterized by a dense network of rivers and streams and receives an annual rainfall of about 2,000 mm. The coastal forest area is characterized by a secondary forest with grassland and small shrub trees. There are few streams and ponds, which usually dry out during the dry season. Tree cover in

the area is, however, rapidly decreasing as a result of farming and economic activities and has led to deforestation. Average rainfall ranges between 350 and 1,200 mm annually.

Biting and indoor resting adult female *An. funestus* mosquitoes were collected from the different ecological areas in Ghana using human landing catches and pyrethrum spray collections, respectively (WHO 1975). A total of 32 human-nights of human landing catches (LC) were carried out indoors and outdoors of rooms every month by two teams of four collectors. The wet season and dry season collections were done twice weekly in the month of June, 2001 and in January, 2002, respectively. Mosquitoes attempting to bite the collectors were collected and kept in paper cups covered by mesh screen. Collections were made from 18:00 to 06:00 and the collectors were rotated between indoors and outdoors after each hour of collection to compensate for differences in the individual mosquito attraction or repulsion. The study received ethical approval from the Noguchi Institutional Review Board. The mosquito collectors were offered treatment in case of any infection. The pyrethrum spray collection (PSC) was carried out between 06:00–09:00. This involved spreading white sheets on the floor of rooms and after closing all the windows, spraying with pyrethrum insecticide. After about 10 min, the white sheets were inspected and all mosquitoes knocked down or dead were collected.

Mosquito processing

Mosquitoes captured by landing catches were anaesthetized with ether, sorted, and morphologically identified into species (Gilles and de Meillon 1968) and counted. All the anophelines caught were stored in 1.5 ml Eppendorf snap vials with desiccants for further laboratory analysis. Heads and thoraces of a sample of *An. funestus* were removed and tested for the presence of circumsporozoite antigens (CS) of *P. falciparum* using the enzyme-linked immunosorbent assay (ELISA) described by Wirtz et al. 1987. The abdomen of blood-fed *An. funestus* individuals were analyzed to identify the source of blood meal using the ELISA method (Beier et al. 1988).

Genomic DNA was extracted from the legs of individual *An. funestus* using the method described by Scott et al. 1993. The extracted DNA was used as a template for PCR. The DNA was analyzed to identify the species of *An. funestus* using the cocktail PCR protocol and primers described by Koekemoer et al. (2002). PCR products were electrophoresed in 2.5% agarose gel, stained with ethidium bromide, and visualized on a UV-light.

RESULTS

Species composition of *An. funestus* group

A total of 10,254 *Anopheles* mosquitoes was collected from the three ecological areas. Of the total, *An. funestus* constituted 53.6% (5,496), followed by *An. gambiae* s.l. at 46.1% (4,725), and *An. rufipes*, 0.3% (33). The Sahel savannah area yielded 80% (4,397) of the total number of *An. funestus* captured by both landing and pyrethrum spray methods, followed by 10.8% (594) from the coastal forest and 9.2% (505) from the forest area. Indoor biting collections from all the ecological zones made up 61.1% (767) of the collections while outdoor collections constituted 38.9% (490).

Out of 812 (comprising 281 from Sahel savannah area, 263 from forest, and 268 from coastal forest) *An. funestus* mosquitoes analyzed, PCR amplification of the genomic DNA confirmed 98.3% (798) of the morphologically identified *An. funestus* as belonging to the *An. funestus* group. Further amplification and analysis of the ribosomal DNA showed that *An. funestus s.s.* constituted more than 90.6% (711) of all the biting and indoor resting catches in all the study areas while *An. leesoni* made up 9.4% (74). *An. funestus s.s.* constituted more than 95% and 81% of the LC and PSC, respectively, in all the three ecological areas (Table 1). *An. leesoni* also formed less than 5% and 8% of the LC and PSC, respectively (Table 1).

Feeding behavior

Biting rates of 13.0, 3.3, and 5.5 bites/human/night were estimated for *An. funestus s.s* in the Sahel savannah, forest, and coastal forest zones, respectively. The pattern of indoor and outdoor night-biting in the Sahel savannah area showed that biting rose progressively through the night and peaked during 05:00-06:00 GMT (Figure 1A). Two major peaks were observed in the forest area at 22:00-23:00 GMT and 24:00-01:00 GMT (Figure 1B) and a similar trend was observed in the coastal forest area (Figure 1C).

Sporozoite infectivity and blood meal source

A total of 812 *An. funestus* specimen from the three ecological areas were assayed using ELISA for the presence of *P. falciparum* circumsporozoite proteins. A total of 62 (7.6%) specimens of the *An. funestus* group tested positive for circumsporozoite proteins. Further analysis of the 62 positive samples by PCRs showed that all were *An. funestus* s.s. None of the *An. leesoni* tested positive for circumsporozoite proteins. A *P. falciparum* sporozoite rate of 5.0% (31/623) was estimated for *An. funestus* s.s. in the Sahel

Table 1. Percentage of *An. funestus* individuals collected from Sahel savannah, forest, and coastal forest areas by different collection methods.

Ecological site/Community	Night Biting Catches			Pyrethrum Spray Collection Catches		
	No. examined	% *An. funestus* s.s.	% *An. leesoni*	No. examined	% *An. funestus* s.s.	% *An. leesoni*
Sahel savannah/ Navrongo	140	97.5	2.5	141	91.6	8.4
Forest/Ejura	130	95.4	4.6	133	95.7	4.3
Coastal forest/ Dodowa	134	98.5	1.5	134	97.3	2.7
Total	404			408		

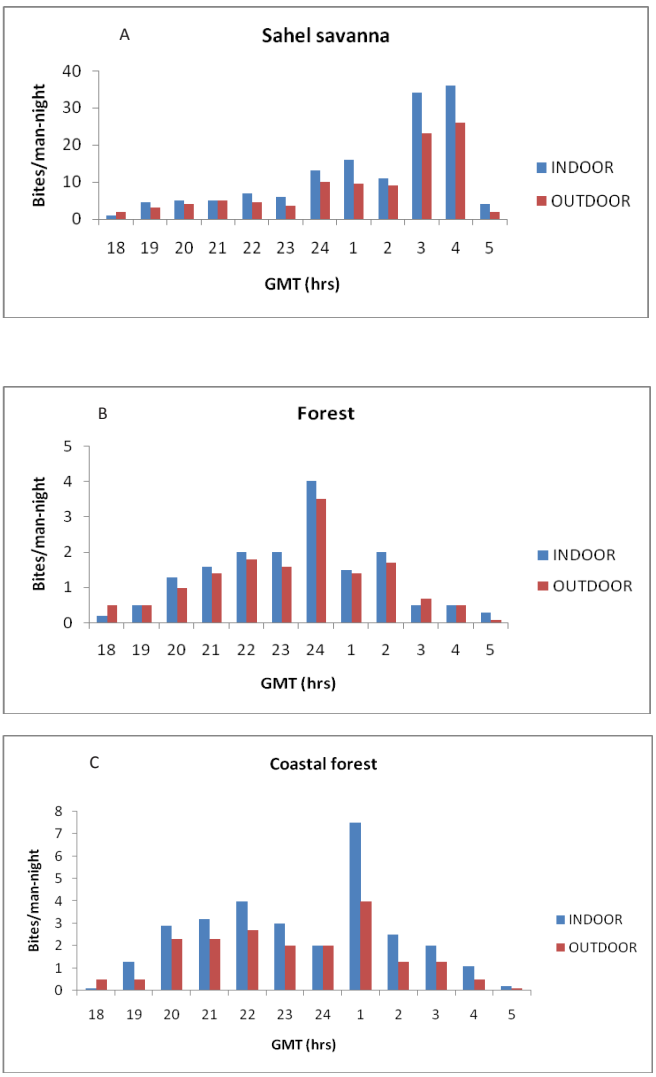


Figure 1. Human night biting pattern of *An. funestus* s.s. captured from A) Sahel savannah, B) forest, and C) coastal forest areas.

savannah area compared to 4.1% (13/318) and 4.7% (18/381) in the forest and coastal forest areas, respectively. A significantly higher (P<0.05) sporozoite rate of 5.5% (23/ 416) was estimated during the wet season than during the dry season, 3.9% (8/207). Analysis of 293 blood meals of *An. funestus* s.s. by ELISA showed that a high proportion (over 80%) of the *An. funestus* s.s. had fed on human blood. The human blood indices of *An. leesoni* in all the three ecological zones were relatively low (4-11%). Most of them had fed on animals (Table 2).

Entomological inoculation rates

The average entomological inoculation rate (EIR) for *An. funestus* s.s. was significantly (P<0.05) higher (0.7 infective/man/night) in Sahel savannah than the forest (0.1) and coastal forest (0.3) areas (Figure 2). Most (86%) of the infective bites occurred indoors in all the study areas. A significantly higher proportion (87%) of the transmission by *An. funestus s.s.* occurred in the wet season than the dry season (Figure 2).

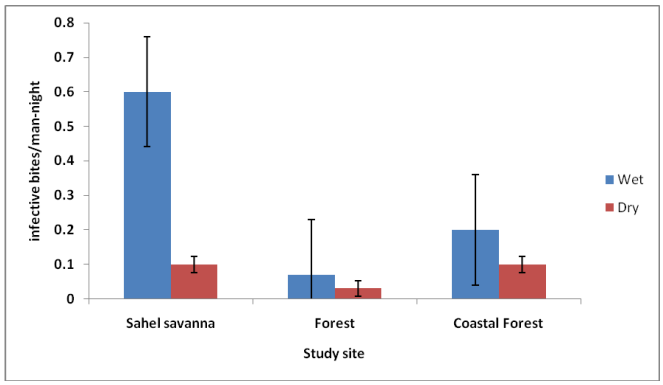


Figure 2. The entomological inoculation rates (EIR, infective bites per human-night) of *An. funestus s.s.* captured from the Sahel savannah, forest, and coastal forest areas in the wet and dry seasons.

DISCUSSION

Members of the *Anopheles funestus* group represent important malaria vectors in Ghana (Appawu et al. 2001a, 2004, Coetzee et al. 2006). The vectorial and behavioral variations found within this species group are the major reasons why accurate identification is needed to understand malaria transmission patterns (Gilles and de Meillon 1968, Koekemoer et al. 1999). The study found that there was sympatry of *An. funestus* s.s. with *An. leesoni* as members of the *An. funestus* group in the three ecological areas. Although *An. gambiae* s.l. was collected from all the study areas, they were not the focus of this investigation and therefore were not included in the current analysis. *An. funestus* s.s. was predominant in the Sahel savannah area compared to the other two sites. The relatively high biting rates of the species in the Sahel savannah area may be due to the presence of an irrigation scheme in the area which provided breeding sites for mosquitoes. In this study, fairly large numbers of *An. funestus* larvae were collected in pockets of water that collected along the irrigated fields. In other areas such as in Madagascar, *An. funestus* has been found to thrive in paddy fields (Gilles and de Meillon 1968).

The presence of *An. funestus* in the forest and coastal areas of Ghana has also been documented (Appawu et al. 2001a, 2004, Coetzee et al. 2006). Previously, it has been suggested that *An. funestus* is rare or absent in coastal areas because the larvae or immature stages cannot tolerate saline waters (Hamon and Coz 1966). There was variation in the pattern of night biting for the species in the three ecological areas. The biting in the Sahel savannah, contrary to what was observed in the same area for *An. gambiae* (Appawu et al. 2001a), increased steadily throughout the night and peaked at dawn (04:00-06:00 GMT) compared to much earlier peaks observed in the forest and coastal forest areas. The pattern of feeding behavior (feeding at dawn) observed in the Sahel savannah area may be due to the continuous use of pyrethroid-based long lasting insecticide treated nets (LLINs) in the area. Compared to the other ecological areas, LLINs coverage in the Sahel savannah area has increased from 4% in 1996 (Binka et al. 1996) to 47% in 2008 and to as low as 20% in the Greater Accra region. Results of a study in Benin (Moiroux et al. 2012)

Table 2. Analysis of blood meal source of members of the *An. funestus* group captured resting indoors in the Sahel savannah, forest, and coastal forest areas.

Ecological Site/ Community	*An. funestus* s.s.				*An. leesoni*			
	No. examined	% Human-fed	% Animal-fed	% Human-animal fed	No. examined	% Human-fed	% Animal-fed	% Human-animal fed
Sahel Savannah/Navrongo	89	91.0	7.9	1.1	18	11.1	83.3	5.6
Forest/Ejura	64	82.8	15.6	1.6	25	4.0	92.0	4.0
Coastal forest/Dodowa	76	96.1	2.6	1.3	21	0.0	95.2	4.8
Total	229	90.4	8.3	1.3	64	4.7	90.6	4.7

showed that after three years of implementation of LLIN at the community level, *An. funestus* bit later during the night (almost at dawn) and more frequently outdoors, compared with the baseline survey. In Kenya, the introduction of LLINs affected the ecological and biological features of the main malaria vector, *An. gambiae* (Bayoh et al. 2010). Personal protection measures such as the use of repellents would be recommended to prevent bites at dawn when most inhabitants are awake and out of the bed nets.

Anopheles funestus, An. rivulorum, and *An. leesoni* have a wide geographic distribution, extending throughout sub-Saharan Africa, and studies have shown that they can also be found in sympatry with other members of the group (Gilles and de Meillon 1968). No *An. rivulorum* was found in any of the three ecological sites. *An. funestus s.s.* is the most anthropophilic and endophilic member of the *An. funestus* group (Gillies and Coetzee 1987, Gilles and de Meillon 1968), and therefore it is not surprising that they constituted more than 90% of the night biting and indoor resting population from all three ecological areas. Although *An. leesoni* is known to be zoophilic and exophilic (Gilles and de Meillon 1968), between 1.5% and 8.4% of the species were among the indoor night biting and indoor resting catches from the three study sites. This is not surprising, because in South Africa, the species was found resting indoors in sympatry with three others of the group (Hargreaves et al. 2002). Again, the population of *An. leesoni* in the night landing catches was relatively small, and the finding is of interest because this species is known not to bite humans and hence its occurrence in night biting collections may indicate a change in its biting behavior, although this needs to be further investigated. On the other hand, *An. funestus s.s.* from all three ecological areas were found to be highly anthropophilic and endophilic with a human blood index of over 90%. This species is known to be the most anthropophilic and endophilic member of the group and can exist in sympatry with other members (Gilles and de Meillon 1968, Gilles and Coetzee 1987). All the CS-ELISA sporozoite positive samples were *An. funestus s.s.* and this suggests that the species may be the only vector species within the group in the three ecological areas, although a much larger sample size may give more conclusive results about its role in malaria transmission.

In Ghana, *An. funestus* s.s. has been found in Obuasi, a forested mining area in Ghana with *P. falciparum* infection rates lower than in *An. gambiae* (Coetzee et al. 2006). The current study found that *Anopheles funestus* s.s. and *An. leesoni* are sympatric in the three ecological areas, with the former playing an important role as a malaria vector in Ghana. In Obuasi, *P. falciparum* infection rates ranged from 4.5% to 8.6% in *An. gambiae* and 1.81% to 8.06% in *An. funestus* (Coetzee et al. 2006). The current study estimated overall *P. falciparum* parasite rates of 7.6% for *An. funestus* s.s. and the rates ranged from 4.1% to 5.0% in all the areas sampled. The Sahel savannah area recorded the highest *P. faciparum* sporozoite rate. This rate was lower than what was estimated for the same species by Appawu in the area in 2001 (Appawu et al. 2004). In other West African countries, parasite rates of between 2.6% and 3.3% in *An. funestus* have been recorded in Senegal (Dia et al. 2003, Fontenille et al. 1997), between 2.8% and 14.6% in Burkina Faso (Costantini et al. 1999) and around 5% in Cameroon (Antonio-Nkondjio et al. 2002). Malaria transmission estimated as annual entomological inoculation rate (EIR) by *An. funestus* s.s. was higher in the Sahel savannah area than the forest and coastal forest ecological areas (255.5 vs 36.5 vs 109.5 infective bites/man/year), respectively. In Burkina Faso, *An. funestus* s.s was found to co-exist with *An. leesoni* in a humid savannah area but malaria transmission was carried out mainly by *An. funestus* s.s (Dabire et al. 2007). In a forest-savannah transition area in Cameroon, *An. funestus* was responsible for 88% of the total malaria transmission, with a *P. falciparum* circumsporozoite rate of 6.8% and an anthropophilic rate of 99.3% (Cohuet et al. 2004).

An. leesoni was present in all the ecological sites in relatively fewer numbers but was not implicated in malaria transmission in these areas. However, the fact that a few were caught biting humans and contained human blood makes it a potential vector in the areas. The low human blood index estimated for the species may be due to the zoophilic behavior of the species as has been found in other areas where they occur (Gilles and de Meillon 1968, Gillies and Coetzee 1987). We conclude that *An. funestus* s.s. is an important vector of malaria in the areas studied and it will be beneficial to extend the study to other areas to determine the extent of variation in the country. This information will be useful in the future formulation of vector control strategies and will also help to understand disease transmission patterns in the country.

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